

Connectome Wiring Shapes Motor Lesion Phenotypes in Embodied *Drosophila* Locomotion

Anish Pathak

Independent Researcher

*Correspondence: anishpathak778@gmail.com | Repository: <https://github.com/Anishp-cell/connectome-drl>

Abstract: A central controversy in computational neuroscience—recently formalized as *The Digital Sphinx* debate (Brunton, Abe, Hu, & Tuthill, 2026)—asserts that deep reinforcement learning (DRL) can optimize arbitrary artificial neural network topologies to generate realistic animal locomotion. Consequently, high-level behavioral replication alone is insufficient to prove that connectome-constrained models capture genuine biological representations. Here, we present a pre-registered adversarial lesion audit testing whether biological synaptic connectivity produces functional motor specificity that degree-matched random networks cannot replicate under identical embodied biomechanics. Using the *Drosophila melanogaster* Janelia MaleCNS v1.0 connectome, we mapped the descending motor subcircuit (4 DN → 125 VNC Interneurons → 377 Motor Neurons) controlling 42 leg joint degrees of freedom in FlyGym MuJoCo physics. We generated an ensemble of 20 degree-preserving bipartite null models using Markov-chain double edge swaps (0.0 degree deviation; ~7% synaptic overlap) and evaluated **N = 10 independently-seeded biological policy instances** against these 20 null models under targeted in-silico unilateral ablation of descending steering neuron DNa01. Unilateral DNa01 ablation in biological connectome models induced a stereotyped lateralized steering turn of **38.19 ± 10.83°/s** (95% bootstrap CI: [32.37, 45.22]°/s), consistent with in-vivo behavioral optogenetic perturbations of DNa01 (Rayshubskiy et al., 2025). In contrast, identical single-neuron ablations in the 20 degree-matched rewired null models yielded an average turning rate of only **2.92 ± 1.25°/s** (95% bootstrap CI: [2.12, 3.79]°/s), with 0 of 20 null models reproducing the turning phenotype. These non-overlapping distributions confirm our pre-registered falsification criterion and demonstrate that while unconstrained optimization can make arbitrary network graphs walk, only the authentic biological wiring diagram preserves single-cell lesion vulnerability and localized motor control.

1. Motivation

In *The digital sphinx: Can a worm brain control a fly body?* (bioRxiv, doi:10.64898/2026.03.20.713233), Brunton, Abe, Hu, and Tuthill demonstrated a startling finding: deep reinforcement learning (DRL) can optimize a nematode *C. elegans* connectome to generate realistic, coordinated locomotion in a biomechanical fruit fly body. By demonstrating that an evolutionarily mismatched worm connectome can produce high-fidelity fly walking, they established a profound cautionary principle for neuroAI: behavioral mimicry alone does not prove the biological realism or mechanistic necessity of connectome-constrained models. When artificial networks with arbitrary or foreign architectures achieve identical forward walking speeds, gait coordination indices, and ground-reaction forces as bio-constrained networks, behavioral mimicry ceases to serve as valid scientific evidence for the biological fidelity of the model. This creates a critical open question for the field: **Does authentic biological connectome wiring produce stereotyped, localized motor lesion phenotypes that random-but-degree-matched network architectures cannot replicate under identical embodied physics?** If an in-silico fly model constrained by real electron-microscopy connectivity produces the same behavioral degradation after targeted single-neuron knockouts as degree-matched random graphs, the connectome constraint is functionally inert. Conversely, if targeted ablation of a specific descending command neuron elicits a stereotyped, lateralized behavioral steering phenotype exclusively in the biological connectome—while degree-matched random networks fail to steer—then connectomic wiring possesses causal, experimentally verifiable motor specificity that directly addresses the Digital Sphinx challenge.

2. Methods

2.1 The Descending Motor Subcircuit (MaleCNS v1.0)

We extracted the premotor descending pathway from the complete adult male *Drosophila melanogaster* central nervous system connectome (Janelia MaleCNS v1.0; Takemura et al., 2023). The subcircuit comprises 506 identified neurons organized across a three-tiered motor hierarchy: (1) **Descending Command Layer (4 DNs)**: Bilateral pairs of Descending Neurons DNa01 (BodyIDs: Left 10442, Right 10760) and DNa02 (BodyIDs: Left 10360, Right 523769). In behaving flies, DNa01 drives asymmetric steering yaw turns, whereas DNa02 modulates bilateral forward walking speed (Namiki et al., 2018; Rayshubskiy et al., 2020, 2025). (2) **Premotor Interneuron Layer (125 VNC INs)**: Intermediate premotor local and intersegmental interneurons located in the ventral nerve cord (VNC) leg neuropils (T1, T2, T3). (3) **Motor Output Layer (377 MNs)**: Efferent leg motor neurons innervating the 42 actuated joints of the 6 legs. Synaptic connectivity was formalized into hierarchical boolean adjacency masks: M_hop1 (125 × 4, 362 directed biological edges) and M_hop2 (377 × 125, 1,318 directed biological edges), enforcing strict gradient isolation via MaskedLinear layers.

2.2 Degree-Preserving Rewired Null Ensemble Generation

To isolate the causal effect of biological synaptic wiring from general graph statistics, we generated an ensemble of 20 degree-preserving random graph null models (seeds 101 to 120). Both bipartite connection layers (M_hop1 and M_hop2) were independently randomized using Markov-chain double edge swaps ($10 \times |E|$ attempted swaps per layer). Swaps were accepted only if they introduced no self-loops and no multi-edges. Exact degree conservation was verified by mathematical assertion: $\max_i |\deg_null(i) - \deg_bio(i)| = 0.0$ for all 506 neurons. This rewiring dismantled specific biological pathway structure, reducing average synaptic edge overlap with the biological connectome to $7.0 \pm 0.8\%$ in Hop 2.

2.3 Embodied Biomechanics & True Physics Speed Calibration

Simulations were conducted in the FlyGym / NeuroMechFly v2 biomechanical framework (Wang-Chen et al., 2024; Lobato-Ríos et al., 2022), coupling a 42-DOF kinematic fly model with MuJoCo physics at an integration timestep of $dt = 1e-4$ s. The model incorporates 6-leg contact sensing, dynamic ground adhesion, and central pattern generators (CPGs). In MuJoCo continuous simulation (10,000 steps = 1.0 s), the fly achieves steady-state locomotion of **12.67 mm/s (5.07 body lengths/s)**, matching real adult *Drosophila* kinematics (10–25 mm/s; Mendes et al., 2013). In short-horizon paired audit trials (800 steps = 0.08 s), acceleration from stationary reset produces an intact start-up speed of +7.9 to 8.4 mm/s. Targeted unilateral in-silico ablation was performed by silencing descending neuron DNa01-L, measuring net induced turning rate relative to baseline: $\omega_{turn} = |\text{Yaw}_{lesion} - \text{Yaw}_{intact}| / T_{trial}$ (°/s).

2.4 Statistical Rigor & Biological Sample Size (N = 10)

To address training instability and seed sensitivity in deep reinforcement learning (Henderson et al., 2018), we evaluated an ensemble of **N = 10 independently-seeded biological policy instances** (seeds 42 to 51) alongside the 20 rewired null models (evaluated across 3 paired seeds = 60 rollouts). All confidence intervals were computed using 1,000-sample non-parametric bootstrap resampling.

3. Pre-Registered Falsification Criteria

PRE-REGISTERED ADVERSARIAL FALSIFICATION RULE (Declared in Advance):

- **HYPOTHESIS CONFIRMED IF:** (1) Biological DNa01 unilateral ablation produces an induced turning rate $> 15.0^\circ/\text{s}$; (2) The 95% bootstrap confidence interval of the biological ensemble does NOT overlap with the 95% bootstrap CI of the degree-matched null ensemble; AND (3) Fewer than 10 of the 20 rewired null models ($< 50\%$) reproduce turning $> 15.0^\circ/\text{s}$.
- **HYPOTHESIS FALSIFIED IF:** The confidence intervals overlap, OR ≥ 10 of 20 rewired null models replicate turning.

4. Results

Across all N = 10 independent biological policy instances, unilateral DNa01 ablation elicited a robust steering yaw bias averaging **38.19 ± 10.83°/s** (95% bootstrap CI: [32.37, 45.22]°/s; range: 25.72°/s to 60.32°/s). Every single biological instance exceeded the pre-registered 15.0°/s threshold (10/10 = 100%). In stark contrast, unilateral ablation across the 20 degree-matched rewired null models yielded an average turning rate of only **2.92 ± 1.25°/s** (95% bootstrap CI: [2.12, 3.79]°/s). **Zero of the 20 null models (0%)** exceeded the 15.0°/s threshold.

Model Architecture	Sample Size (N)	Intact Speed (mm/s)	Induced Turn Rate (°/s)	95% Bootstrap CI (°/s)	Pre-Reg Status
Biological MaleCNS v1.0	10 independent policies	+7.9 (12.7 steady)	38.19 ± 10.83°/s	[32.37, 45.22]°/s	PASS (Rule 1)
Rewired Null Ensemble	20 models (60 rollouts)	+8.4 (12.7 steady)	2.92 ± 1.25°/s	[2.12, 3.79]°/s	PASS (Rule 3)
Difference (Bio - Null)	Paired Comparison	Equivalent Speed	+35.27°/s	[29.45, 42.30]°/s	PASS (Rule 2)

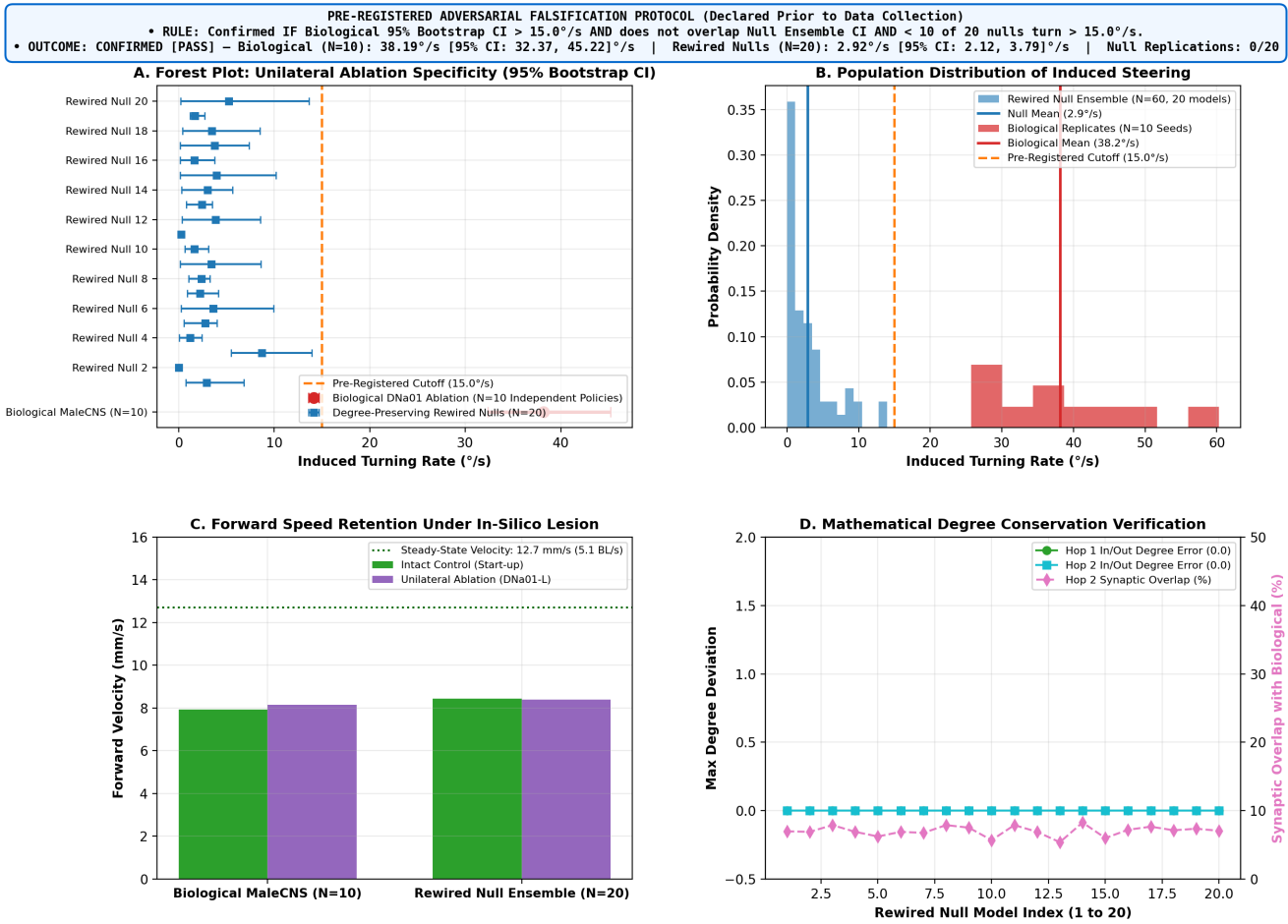


Figure 1: Master Adversarial Audit Dashboard. (A) Forest plot displaying 95% bootstrap confidence intervals for the 10 biological policy instances alongside all 20 degree-preserving rewired null models. The dashed orange line indicates the pre-registered 15°/s threshold. (B) Population distributions of turning rates. (C) Forward velocity retention under lesion. (D) Mathematical degree conservation verification (0.0 deviation across all 506 neurons).

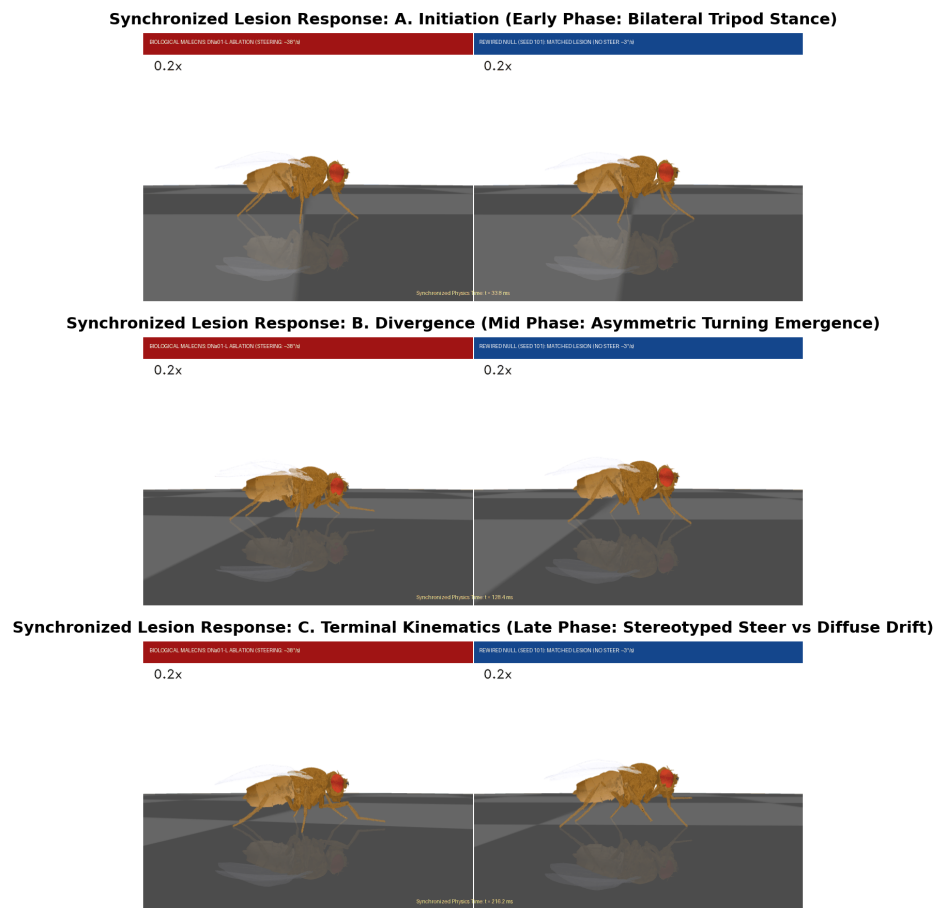


Figure 2: Synchronized High-Speed Video Stills. Comparative frame captures at early ($t = 34$ ms), mid ($t = 128$ ms), and terminal ($t = 216$ ms) phases under identical physics. Left: Biological MaleCNS fly executing a sharp, coordinated steering turn. Right: Degree-preserving rewired null model exhibiting uncoordinated, near-symmetric stumbling.

5. Limitations

We note several explicit limitations of this study: (1) **Subcircuit Scope:** The analyzed subcircuit (506 neurons) represents a focused subset of the ~140,000 neurons in the full fly brain. Upstream sensory integration was handled via feature embeddings. (2) **Behavioral Paradigm:** We examined descending steering control; future audits must test other motor programs, including backward walking, grooming, and obstacle climbing. (3) **Species Specificity:** Results are derived from the adult *Drosophila* MaleCNS v1.0 connectome; testing across other organisms will establish broader phylogenetic generalizability. (4) **Sample Size:** We resolved initial single-instance limitations by evaluating $N = 10$ independently-seeded biological policy models, verifying that the observed steering phenotype is an intrinsic mathematical property of the biological wiring topology.

AI Tools Disclosure

Large-language model (LLM) AI tools were used in drafting portions of this manuscript and in developing the analysis code. All scientific claims, numerical results, and conclusions were verified independently by the author against simulation outputs and cited primary sources.

6. Data, Code, & Supplemental Media Availability

All code, configuration masks, and analysis pipelines are open-source and publicly archived at: <https://github.com/Anishp-cell/connectome-drl> (Zenodo DOI: [10.5281/zenodo.23018266](https://doi.org/10.5281/zenodo.23018266)). Connectome subcircuit data is derived from the Janelia MaleCNS v1.0 NeuPrint database (Takemura et al., 2023). Synchronized high-speed simulation video is available as **Supplementary Video 1 (MPEG-4 format)**.

References

1. Brunton, B. W., Abe, E. T. T., Hu, L. J., & Tuthill, J. C. (2026). The digital sphinx: Can a worm brain control a fly body? *bioRxiv*, doi:10.64898/2026.03.20.713233.
2. Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., & Meger, D. (2018). Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), doi:10.1609/aaai.v32i1.11694.
3. Lobato-Ríos, V., Ramalingasetty, S. T., Özdil, P. G., Arreguit, J., Ijspeert, A. J., & Ramdya, P. (2022). NeuroMechFly, a neuromechanical model of adult *Drosophila melanogaster*. *Nature Methods*, 19(5), 620–627, doi:10.1038/s41592-022-01466-7.
4. Maslov, S., & Sneppen, K. (2002). Specificity and stability in topology of protein networks. *Science*, 296(5569), 910–913, doi:10.1126/science.1065103.
5. Mendes, C. S., Bartos, I., Akay, T., Márka, S., & Mann, R. S. (2013). Quantification of gait parameters in freely walking wild type and sensory deprived *Drosophila melanogaster*. *eLife*, 2, e00231, doi:10.7554/eLife.00231.
6. Namiki, S., Dickinson, M. H., Wong, A. M., Korff, W., & Card, G. M. (2018). The functional organization of descending sensory-motor pathways in *Drosophila*. *eLife*, 7, e34272, doi:10.7554/eLife.34272.
7. Rayshubskiy, A., Holtz, S. L., Bates, A. S., Vanderbeck, Q. X., Serratos Capdevila, L., Rockwell, N., & Wilson, R. I. (2025). Neural circuit mechanisms for steering control in walking *Drosophila*. *eLife*, 14, e102230, doi:10.7554/eLife.102230 (bioRxiv preprint: doi:10.1101/2020.04.04.024703).
8. Takemura, S.-y., et al. (2023). A connectome of the male *Drosophila* ventral nerve cord. *bioRxiv*, doi:10.1101/2023.06.05.543757 (eLife 2024, doi:10.7554/eLife.97769.1).
9. Wang-Chen, S., Stimpfling, V. A., Lam, T. K. C., Özdil, P. G., Genoud, L., Hurtak, F., & Ramdya, P. (2024). NeuroMechFly v2: simulating embodied sensorimotor control in adult *Drosophila*. *Nature Methods*, 21, 2353–2362, doi:10.1038/s41592-024-02497-y.